

Ashutosh Naik

Software Engineer · Backend Systems · Event-Driven Infrastructure · Applied AI
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EXPERIENCE

Senior Software Engineer

July 2024 – Present

Nomura Holdings America, Inc.

New York, NY

- Drove the migration of Nomura's credit ETL system (RETL) from Prefect 1.x to 2.x as part of a broader event-based processing (EBP) transition; refactored tightly coupled, monolithic sequential flows into parallel, modular task subflows and integrated Dask-backed distributed execution, eliminating serial processing bottlenecks across production credit risk data pipelines.
- Architected the "Mario" configuration microservice (FastAPI, PostgreSQL) with Smart Insertion logic to suppress redundant writes and SQLAlchemy connection pooling for concurrent load; integrated Mario into an event-driven architecture via the Stargate hub (Solace PubSub+, Mercury Messaging), enabling asynchronous orchestration between market data providers and downstream calculation engines.
- Led infrastructure-wide migration to AWS Graviton (ARM64); diagnosed and resolved subtle x86→ARM64 integer overflow discrepancies embedded in Basel-compliant risk model codebases through low-level cross-architecture debugging, restoring 100% data parity across production risk systems.
- Built a RAG-based credit reporting tool (LangChain, LLMs) with a natural language query interface over complex financial reports, automating manual research workflows for credit operations teams.
- Engineered a Cloud Lifecycle service automating EC2 shutdown workflows and CNAME management via REST API, reducing infrastructure overhead by 12%; hardened CI/CD pipelines with Podman, Ansible, and GitLab CI/CD for containerized production deployments.

Graduate Student Researcher

Aug. 2022 – April 2024

University of Colorado Boulder

Boulder, CO

- Primary implementer of ShelfAware (IEEE RAL 2025): designed a real-time semantic particle filter fusing depth likelihoods with distributional semantic observations (Jensen-Shannon Divergence over object-class counts) and an inverse semantic proposal mechanism — achieving 96% global localization success vs. 22% MCL / 10% AMCL baselines at 9.6 Hz on consumer laptop hardware.
- Built ShelfHelp's two-stage product detection pipeline (AAMAS 2023): YOLOv5 region proposals + frozen autoencoder with cosine similarity, enabling scalable detection of thousands of SKUs without per-product retraining; formulated an MDP-based manipulation guidance system achieving 90.66% task success with blindfolded users (n=15).
- Designed an autonomous anomaly detection and explanation system fusing trajectory-based spatial analysis with GPT-4V for semantic anomaly interpretation; introduced Jensen-Shannon Divergence for shelf stock fluctuation quantification, extending VLM numerical reasoning capabilities beyond existing baselines.

PROJECTS

ScreenSuggest — Python, ChromaDB, SvelteKit, ONNX Runtime

Apr. 2024

- Built a semantic movie recommendation engine using ChromaDB for vector search over 50k+ titles; quantized the 'all-mpnet-base-v2' embedding model to ONNX, reducing memory footprint and accelerating query latency.

CloudBoard — Go, Redis, GCP Cloud Run, WebSockets, JWT

Jun. 2023

- Engineered real-time cross-device clipboard synchronization via WebSocket sessions with JWT authentication and Gin REST APIs; deployed on GCP Cloud Run for high availability.

SKILLS

Languages: Python, Go, C/C++, SQL, JavaScript, Bash

Frameworks & Libraries: FastAPI, SQLAlchemy, Prefect 2.x, Dask, LangChain, PyTorch, OpenCV, Pandas, NumPy

Data & Storage: Apache Iceberg, Dremio, MinIO, PostgreSQL, Redis, MSSQL Server

Infrastructure & Cloud: AWS (EC2, Graviton/ARM64), GCP (Cloud Run), Docker, Kubernetes, Podman, Terraform

Tools & CI/CD: GitLab CI/CD, Ansible, Solace PubSub+, Pytest, SonarQube, Nexus

PUBLICATIONS

ShelfAware: Real-Time Semantic Localization in Quasi-Static Environments — IEEE Robotics and Automation Letters (RAL), 2025

ShelfHelp: Empowering Humans to Perform Vision-Independent Manipulation Tasks with a Socially Assistive Robotic Cane — AAMAS 2023

EDUCATION

Master of Science in Computer Science, University of Colorado Boulder — GPA: 4.0 / 4.0

May 2024

Bachelor of Technology in Computer Engineering, University of Mumbai — GPA: 9.77 / 10

May 2022